

ITCS285 Database Management Systems
Second Semester, 2025-2026
Assignment #4

Consider the following relation:

R3 (A, B, C, D, E, F, G, H, I, J, K, L), the primary key is **AKL**

And the set of functional dependencies:

$F = \{ \{A\} \rightarrow \{B\}, \textbf{(1)} \}$

$\{A\} \rightarrow \{D\}, \textbf{(2)}$

$\{A,E\} \rightarrow \{G\}, \textbf{(3)}$

$\{B\} \rightarrow \{C\}, \textbf{(4)}$

$\{D\} \rightarrow \{C\}, \textbf{(5)}$

$\{E\} \rightarrow \{F\}, \textbf{(6)}$

$\{E\} \rightarrow \{H\}, \textbf{(7)}$

$\{H\} \rightarrow \{I\}, \textbf{(8)}$

$\{J,K,L\} \rightarrow \{E\} \textbf{(9)}$

1. Normalize to 2NF
2. Normalize to 3NF
3. Normalize to BCNF